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PROJECTS

INTRODUCTION

In this course you will be required to prepare and submit three written project assignments. The projects are intended to synthesize your understanding of the course content, presenting you with challenging, stimulating, and thought-provoking work. They are designed to be practical exercises encouraging you to apply the concepts covered in this course. These projects are also worth a significant portion of your final course grade: 20% for Project 1, 20% for Project 2, and 10% for Project 3. You should note that corrected Projects 1 and 2 will make up a substantial portion of the Case Study Examination as well.

Below is a summary of the projects to be submitted in this course:

- Project 1: Modeling Land Value and Cost Case Study
- Project 2: Sales-Based Modeling Case Study
- Project 3: Income Modeling Case Study

Projects are to be submitted online. Instructions for submitting your projects can be found on the Course Resources webpage. The project due dates can be found on the Course Resources webpage under “Important Dates and Deadlines”. Because the projects are in-depth and time-consuming, it is recommended you look at these projects well before the due dates in order to determine how much time you personally think you may need to complete the projects. A “Tips” document is provided under the Online Readings for the Projects. It contains some useful information that should assist in completing the Projects. It is highly recommended that students print out the “Tips” document and refer to it while completing their Projects.

Expectations for Your Projects

In general, your reports are expected to provide the level of detail that the intended reader would require in a practical context. This means that you need to report in a clear manner any information that will be helpful to the reader/client in making their decision. Therefore, your report must cover all pertinent information, be well-organized, and professionally presented.

You should write your report at a level understandable by an intelligent reader who is generally informed about real estate but is not an expert on statistical topics. You must write your report specifically to address the information needs of this person so that they can understand and apply your work. It is your job to focus the reader and provide him/her with the relevant information they need. Submitting an excessively long report to a client is likely to be similarly inappropriate to a report that is much too short. As such, brevity would be considered a positive point by the reviewer of your report. You will need to exercise some judgement as to what is not relevant and does not need to be included. If you are in doubt, you may wish to contact your course tutor or your fellow students on the course discussion forum in order to discuss what they are doing and get their advice.

Finally, students are also reminded that, in addition to a practical demonstration of real estate analysis, this is also an examination of the underlying theory. You must ensure that you provide sufficient explanation of the process and procedures in order to demonstrate your knowledge of the area.

Report Specifications

No project in this course should be longer than 25 pages of report detail, with additional supporting documents in appendices. In order not to overwhelm the reader, the supporting documentation in the appendices should be limited to that which helps you to build your case; in other words, you should not include every single piece of information you produce, but only that which you believe is relevant to the reader of the report.

A very well-written and organized report could require significantly fewer pages than the maximum and, as mentioned earlier, this brevity would be considered a positive point by the reviewer of your report. Under no circumstances should you need more than the stated maximum number of pages for your report. If your submission exceeds these limits, EITHER: 1) your assignment will be unmarked and you will receive a request for you to resubmit your

assignment within the page length requirements; OR 2) the additional pages will not be read by the marker and marks will be deducted for their inclusion.

Below are some specific guidelines for preparing and submitting your projects:

- font of 11 point or larger, 12 point preferred;
- preferred line spacing is 1.5;
- minimum 1-inch margins;
- pages numbered consecutively in the top right corner, separately numbered for the report body and appendix; and,
- all sources of information relied upon in producing your submission should be carefully documented in a bibliography or list of references (references used should be acknowledged).

Keep in mind that your assignment submission is expected to be logically organized and look professional. There will be marks allocated for presentation.

Submitting Your Projects

Projects must be uploaded to an online plagiarism checking website. The instructions on how to submit projects is available on the Course Resources webpage. Any ancillary files (such as: MS-Excel workbooks, IBM SPSS syntax files, MS-Word documents) that are necessary to support your report should be sent directly to your tutor by email – the email address for your tutor can be found under “Tutorial Assistance” on your course webpage.

If your tutor is unable to open any of your file attachment(s), then the tutor will contact you to work out a solution.

Guidelines for Submitting Ancillary Files by Email

- Ensure that you specify the course, project number, last name, and student number in the subject line of your email. For example, the subject line should read: “444 Project 2, Smith, 9901011”
- Ensure your attachment file names include your last name and student number.
- Ensure your attachments on a single email note do not exceed a total of 5 MB in size. If your files total more than 5 MB, please split your submission into smaller groups and send separate email notes. The tutor may refuse to accept larger files.

Project Grading and Return

Your tutor will make a concerted effort to grade your project in a timely fashion. The target turn-around time for project marking is two weeks from the date of receipt. However, please keep in mind that the Real Estate Division’s tutors are generally full-time real estate professionals who do tutoring on a part-time basis, and sometimes business pressures make it difficult to attain this target date (e.g., up to one additional week may be required). The tutor has been instructed to give priority to students who are working on-time according to the course schedule. This means that if your project is submitted on-time, it will receive top priority. If your project is submitted late, then your project will be marked after the on-time submissions are graded and other duties are taken care of, such as tutoring.

Staying on-time with projects is particularly important towards the end of the course, when project grading and exam preparation requires a lot of the tutor’s time. You are best advised to keep to the course schedule and to submit your work on-time. Projects 1 and 2 make up a significant part of the Case Study Examination, so you will have a much better chance of being successful on the exam if you have sufficient time to review the marker’s comments and suggestions for your Project 1 and Project 2 submissions.

If you wish to view your project grades on the Assignment History web page, Project 1 is listed as “Assignment 11”, Project 2 is listed as “Assignment 12”, and Project 3 is listed as “Assignment 13”. Please note that there are no answer guides for the projects in this course. The project instructions provide sufficient detail to indicate what is required. When taken together with the tutor’s feedback on your graded assignment, this should give you a clear idea of what was expected on the project and where your work may require improvement. For Projects 1 and 2, it is very important that you make these improvements for your exam submission.

Individual Work Requirements

The projects count for significant credit towards completion of this course, which is an educational requirement in a degree credit program and for accreditation by professional real estate associations. Because of these important consequences, it is vital that students produce and submit work which is entirely their own. Completing the projects in a group is NOT allowed – answers must be produced on an individual basis. The marker of these projects must be able to certify that each individual student has a reasonable understanding of the concepts and techniques employed in statistical analysis in valuation.

For example, each student must:

- describe their own views on the problem, how the background information will affect the problem's solution, and the methodology they will apply in applying this solution;
- create their own mass appraisal model, which applies their personal appraisal assumptions and decisions;
- supply their own individually-produced output to support decisions made and actions taken; and
- provide their own summary of findings and discussion of the validity of the solution.

Any recognized academic misconduct, including copying other students' work or plagiarism from any source, will result in academic consequences as specified in the "Program Regulations" section of the Real Estate Division Student Handbook.

PROJECT 1: MODELING LAND VALUE AND COST CASE STUDY

See page **Projects.1** for details on project options and general expectations. The two projects together will form a comprehensive analysis.

Total Marks for Project 1: 20% of Course Grade



INTRODUCTION

The projects in this course require students to create valuation models based on a database of property sales. Project 1 focuses on modeling for land value and market value under the cost approach. Project 2 focuses on sales-based modeling. Both Projects will make up a significant portion of the Case Study Examination for BUSI 444.

There are several learning outcomes for Project 1:

- Test, develop, and apply time adjustments;
- Develop land values using a land residual method, test the values for accuracy and consistency using the unit value method, and make any necessary adjustments;
- Develop a market-based depreciation factor;
- Develop depreciated building values based on cost data, test the values for accuracy and consistency, and make any necessary adjustments;
- Develop market values using a market-adjusted cost model, test the values for accuracy and consistency, and make any necessary adjustments; and
- Critically evaluate model quality and the process used in developing these values.

Length of Report

Your report must describe the processes you used, demonstrating your understanding of mass appraisal theory and knowledge of how to develop a sound, practical valuation model. However, your submission should be presented in the format of an actual report as much as possible. The only statistics that should be included in the body of the report are those necessary to prove the validity of the opinion being expressed. The full detail graphs and tables should only be included as appendices to the main report. For example, if the mean and standard deviation obtained from a descriptive statistics report are included in your report, these should be followed by a reference to the full report (i.e., “Full details of this report may be found in the descriptive statistics report contained on page 3 of Appendix A”).

Your report should contain no more than 25 pages of report detail and will be supported by an appendix that contains all statistical information you relied on to justify any decisions or reach any conclusions. The appendix should be organized and indexed and may contain, on average, approximately 15 to 30 pages. Note that these are only rough limits and that you should provide all relevant information, i.e., anything which you used to substantiate decisions or adjustments. Note also that a very well-written and organized report could require significantly less than these numbers of pages.

The table below summarizes the suggested size of submissions and the maximum allowable pages. Under no circumstances should you need more than the stated maximum number of pages for your report. If your submission exceeds these limits, EITHER: 1) your assignment will be unmarked and you will receive a request for you to resubmit your assignment within the page length requirements; OR 2) the additional pages will not be read by the marker and marks will be deducted for their inclusion.

	Report	Appendices
Suggested	10 to 15 pages	15 pages
Maximum	25 pages	30 pages

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PROJECT 1, continued



Helpful Hint

For statistical output, be careful to **ONLY** provide the charts and tables that support your discussion – **DO NOT** overwhelm the marker with pages and pages of irrelevant output, as this will be considered a fault and marks deducted. A good rule to follow for statistical output is that you should not have it in your appendix unless you referred to it in your report – if it did not warrant mentioning in your report, then why is it relevant enough to make the marker wade through it?

The marker must be able to follow the logic of your analysis, and so must be able to see the transformations you made. Those working in IBM SPSS must include syntax files showing all transformations made. Ideally, you should provide an electronic copy of your syntax file, and any relevant output files so that the marker can review your work if necessary. Alternately the transformation syntax statements can be included as part of the appendices. It is especially important to include enough ancillary information so that the marker can make comments and suggestions for improving or correcting your Project, because, later in the course, you will use a corrected version of Project 1 as part of your Case Study Examination.

Students are reminded that, in addition to a practical demonstration of mass appraisal processes, this is an examination of the underlying theory. You must ensure that you provide sufficient explanation of the processes and procedures in order to demonstrate your knowledge of sound mass appraisal theory. While solid results are important, a good percentage of your grade is based on the explanation provided.

Project 1 – Land Value and Cost Modeling Case Study

In this project, you must develop and test a market-adjusted cost model to appraise improved residential properties in Marilu City. Marilu City uses a full market value assessment base, so your model values must reflect the full actual value of the fee simple interest in the properties.

Your report must describe your results and the procedures you used to obtain these results. The report should provide the Assessor of Marilu City with a mass appraisal model that could be applied to value all improved residential properties in Marilu City. The valuation date is July 1, 2017 and the assessment roll is to be presented to Marilu City on December 31, 2017. Your report will be presented to the Assessor on November 30, 2017. As a part of this exercise, you are expected to also fully critique the model, the results, and their applicability.

Database Description

The project database is titled “MariluProject1_YEARTERM” and can be found on the BUSI 444 Course Resources web page in the same location as the other database files for the lessons. It is in IBM SPSS format.

The database contains 57 existing variables. There are 400 or more sales included in the database (each term has a slightly different database provided). The description of each variable is available under the “Variable View” tab in SPSS.

Project 1 continues on the following page

PROJECT 1, continued

The Marilu Database

The MariluProject1 database contains the following 57 variables:

1. SALEDATE	16. MBMCLSDV	31. GAR_AREA	46. PIE_SHPE
2. NEIGH	17. BEDROOMS	32. GAR_TYPE	47. SLOPE
3. SALEPRICE	18. BATHS	33. GRMKTVAL	48. ABV_ROAD
4. LANDVAL	19. FAMILYRM	34. GR_RCN	49. SPLITLVL
5. IMPRVAL	20. FIREPLCS	35. OB1MKTVAL	50. CRWLAREA
6. TOTALVAL	21. FLRAREA1	36. LCS_VAL	51. BSMTAREA
7. MBMANCLS	22. FLRAREA2	37. WIDTH	52. UNFINBMT
8. MBYRBLT	23. BSMTFIN	38. DEPTH	53. AREA1_2
9. MBEFFYR	24. MBMKTVAL	39. LOT_SIZE	54. TOTAREA
10. HEATTYPE	25. MB_RCN	40. CONDUIT	55. LINMCDV
11. BASESTE	26. CP_MANCL	41. ALLEY	56. RANDOM
12. GARAGEFL	27. CP_AREA	42. SIDEWALK	57. MONTH
13. CARPTFL	28. CPMKTVAL	43. CULDESAC	
14. MKTRCNLD	29. CP_RCN	44. CORNER	
15. LINMCLS	30. GR_MANCL	45. WOODED	

The data labels for each variable are as follows:

Variable	Name	Description
1	SALEDATE	date of sale
2	NEIGH	sub-neighbourhood
3	SALEPRICE	sale price
4	LANDVAL	land assessed value (previous year)
5	IMPRVAL	improvement assessed value (previous year)
6	TOTALVAL	total assessed value (previous year)
7	MBMANCLS	main building class (quality and storey)
8	MBYRBLT	main building year built
9	MBEFFYR	main building effective year built
10	HEATTYPE	type of heating system
11	BASESTE	basement suite (1=yes)
12	GARAGEFL	garage flag (1=yes)
13	CARPTFL	carport flag (1=yes)
14	MKTRCNLD	market adjusted RCNLD (all buildings)
15	LINMCLS	scaled value of MBMANCLS
16	MBMCLSDV	% deviation from main building quality class
17	BEDROOMS	number of bedrooms

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Project 1 continues on the following page

PROJECT 1, continued

Variable	Name Label	Description
18	BATHS	number of bathrooms
19	FAMILYRM	count of family or recreation rooms
20	FIREPLCS	count of fireplaces
21	FLRAREA1	first floor area
22	FLRAREA2	second floor area
23	BSMTFIN	finished basement area
24	MBMKTVAL	main building market adjusted RCNLD
25	MB_RCN	main building Replacement Cost New
26	CP_MANCL	carport quality class
27	CP_AREA	total carport area that is not part of the basement
28	CPMKTVAL	mkt adjusted RCNLD for carport
29	CP_RCN	carport RCN
30	GR_MANCL	garage quality class
31	GAR_AREA	garage area
32	GAR_TYPE	type of garage
33	GRMKTVAL	market adjusted RCNLD for garage
34	GR_RCN	garage RCN
35	OB1MKTVL	market value estimate of first outbuilding
36	LCS_VAL	land value estimate from Land Calculation System
37	WIDTH	lot width in feet
38	DEPTH	lot depth in feet
39	LOT_SIZE	lot area in square feet
40	CONDUIT	binary – electric, telephone, etc. services in underground conduit
41	ALLEY	binary – rear alley present
42	SIDEWALK	binary – frontage has sidewalk
43	CULDESAC	binary – lot fronts on cul-de-sac
44	CORNER	binary – corner lot
45	WOODED	binary – lot is more wooded than typical lot in NBHD
46	PIE_SHPE	binary – lot is pie shaped
47	SLOPE	binary – sloped lot
48	ABV_ROAD	binary – above road
49	SLITLVL	binary – split level style of construction
50	CRWLAREA	crawl space area
51	BMST AREA	full-height basement area
52	UNFINBMT	unfinished basement area
53	AREA1_2	sum of first and second floor area
54	TOTAREA	total finished area including basement
55	LINMCDV	scaled class plus deviation factor
56	RANDOM	random number used for sort of data
57	MONTH	month in which sale occurred

Project 1 continues on the following page

PROJECT 1, continued

Most of these variables require no explanation other than the description provided. A few variables are elaborated on below:

- SALEDATE (variable #1), the format for variable SALEDATE in this database is DDMMYY. This format makes the extraction of month of sale a little difficult, so the variable MONTH (variable #57) has been provided in this database.
- MBMANCLS (#7 – main building manual class), is a variable that represents two building characteristics: the last digit indicates the number of storeys, while the first digit (or digits) indicates the quality of construction and typical architectural styles related to the time of construction. The quality classes in the Marilu database are 3, 4, 5, 8, 9, 14, 15 and 16. There is no direct link between the number and the quality. For example, the 5 and 8 classes are approximately the same quality, but the 5 class was built before 1930 whereas the 8 class was built more recently. Similarly, the 9 and 14 classes are very similar in quality with the 9 class built before 1960 and the 14 class built more recently. The last digit indicates storey height: a “0” is a 1-storey, a “1” is a 1½-storey, and a “2” is a 2-storey. Therefore, a 140 is a 1-storey, a 141 is a 1½-storey, and a 142 is a 2-storey version of the same general quality house. Architectural styles for each manual class are as follows:

30, 31, 32: All ages – poor

This class provides the minimum in shelter, generally falling far short of building requirements. It meets only basic needs. It is commonly found in rural areas and older urban areas where no building standards were previously enforced.

40, 41, 42: All ages – substandard

This class seldom meets modern building code requirements, although those dwellings built before 1930 are typical of low-cost housing. Generally, the more modern versions are found in suburban or rural areas.

50, 51, 52: Before 1930 – fair

This class provides basic housing at a low to moderate cost for the era in which these dwellings were built. The architectural details, both exterior and interior, were simple. The floor plans were also simple, with relatively large rooms and few passageways and storage cupboards provided. Ceiling height up to ten feet.

80, 81, 82: After 1930 – fair

This class provides basic housing, with low construction cost being the main objective. The “wartime house” of the 1940s era, and the owner-built dwelling with a plain appearance are characteristic of this class of construction. Interior finish is basic throughout, with few, if any, decorative features included.

90, 91, 92: After 1930 – standard

This class provides housing at a moderate cost. It will include a few refinements in limited quantity, such as inexpensive panelling or exterior ornamentation. The class satisfies minimum Canada Mortgage and Housing Corporation (CMHC) standards and is usually built by speculation to meet the basic housing market.

140, 141, 142: After 1960 – modern standard

This class provides standard quality housing with a modern theme. It is often characterized by small quantities of more costly exterior ornamentation and frequently has a cathedral style or basement entry. It meets or exceeds CMHC standards.

145, 146, 147: After 1980 – new standard

This class provides a new standard of quality housing in the post 1980 period. It is characterized by irregular roof and wall lines and may have some special design features such as vaulted or barrel ceilings. Bay windows are a common feature.

150, 151, 152: After 1930 – semi-custom

This class provides housing that may be contract or speculatively built to meet a higher quality housing market. Attractive and durable exterior finishes and well laid out interior space are features of the class. Refinements are limited to better quality pre-manufactured components.

160,161,162: After 1930 – custom

This class is normally architecturally designed and custom built. The home will conform to the site and take full advantage of any site amenities such as view. Material and finishes will be superior quality. There are very few of these homes in the Marilu database.

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PROJECT 1, continued

It is emphasized again that there is no link between the size of the number assigned to the manual class and the quality it represents. While the manual classes in the Marilu database appear to follow a ranking from poor to good, this is only the case because Marilu is a newer city. Some of these other manual classes do not follow this numerical ranking. For example, the quality represented by a 60 manual class is better than an 80 class, a 70 class is better than a 140 class, and a 74 class is better than a 150 class. If these manual classes had been included in the Marilu database, the level of quality represented by the MBMANCLS variable would not follow the same order as the numerical values assigned.

- HEATTYPE (#10), refers to the type of main source of heat. The different types included in the Marilu database are as follows:

0	no heat
1	gas wall furnace
2	gas fired hot air
3	oil fired hot air
4	electric hot air
5 & 6	electric baseboard
7	gas fired hot water
8	oil fired hot water

NOTE: Value labels have been assigned for these heat types.

- MKTRCNLD (#14 – market replacement cost new less depreciation), is the RCNLD (replacement cost new less depreciation) with market adjustments. RCNLD represents a value which has been calculated directly from a cost manual, with only specific cost-related adjustments for time and location. MKTRCNLD is the RCNLD value with adjustments applied for local market conditions.
- MB_RCN (#25 – main building replacement cost new), is the replacement cost new for the main building adjusted for current and local cost factors.
- CP_AREA (#27), is the total area of the carport that is not part of the basement. Because the cost manual used to develop the rates does not add value for areas of the carport or garage that are under a main floor (in the basement) some carport areas will be zero or very small.
- CP_RCN (#29 – carport replacement cost new), is the replacement cost new for carport adjusted for current and local cost factors.
- GAR_AREA (#31), is similar to carport area, except for garages. The same caution applies. Basement garage costing is simply a constant value (that depends on quality class) to include the additional cost of the garage door. The garage area for basement garages includes only that area that extends beyond the building's main floor. Most are zero or very small areas that cannot be used in any calculations.
- GAR_TYPE (#32), this is a code to indicate the following:

0	No garage
1	Attached garage
2	Detached garage
3	Basement garage

NOTE: Carports do not need to be coded by type as no separate costing is provided for attached or detached carports.

PROJECT 1, continued

- GR_RCN (#34 – garage replacement cost new), is the replacement cost new for garage adjusted for current and local cost factors.
- SPLITLVL (#49), is a binary variable in which a one indicates a split-level style of construction, while zero represents standard 1- or 2-storey type construction.
- CRWLAREA (#50), is the area of a crawl space type basement.
- BSMTAREA (#51), is the full height basement area.
- UNFINBMT (#52), is the area of the portion of a standard basement that is not finished. The sum of this variable and BSMTFIN should equal BSMTAREA.
- RANDOM (#56), is a random number between 0 and 1. The Marilu database can be sorted by this random number to allow models to be developed on a randomly drawn portion of the database and tested on another randomly drawn portion of the database.

Cost Data

The table on the following page provides cost information from the standardized cost manual used in this province. The costs are current, but they may require local adjustment.

Item	Categories, Costs, and Notes	
Basement	Constant Area rate Finished area	included in 1-storey constant \$21.25 add \$15.60 per square ft
Crawl space	Constant Area rate	included in 1-storey constant \$14.90
1-storey	Constant 1-storey area rate	if full height basement \$51,500 if crawl \$45,500 \$42.50
2-storey	Constant 1-storey area rate 2-storey area rate	Add \$16,300 to 1-storey rate \$42.50 \$33.40
Heating	Types 1-6 Hot water heat	included in constant first storey: \$4.40 per square ft second storey: add \$3.30 per square ft
Fireplaces		\$3,850.00
Baths (full)		\$4,000.00
Baths (partial)		fraction times the full bath rate
Main Building Manual Class (factors)		
40, 41, 42	0.507	
50, 51, 52	0.767	
80, 81, 82	0.783	
90, 91, 92	0.945	
140, 141, 142	1.000	
145, 146, 147	1.220	
150, 151, 152	1.190	

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PROJECT 1, continued

Item	Categories, Costs, and Notes	
Garage/Carport Manual Class (factors)		
910/911	0.450	
920/921	1.000	
930/931	1.200	
940/941	1.420	
Carport	Constant Area rate	\$5,850.00 \$22.00 per square ft
Attached Garage	Constant Area rate	\$11,600.00 \$30.00 per square ft
Detached Garage	Constant Area rate	\$16,300.00 \$30.00 per square ft
Basement Garage	Lump Sum	\$7,100.00

Modeling Steps

The following steps must be completed in building your land model and market-adjusted cost model:

1. Test data to determine if a time adjustment to the sale price is required.
2. If required, determine the amount of time adjustment needed and the appropriate method to apply it.
3. Develop an appropriate land value using the unit value method on the land residual.
4. Ensure land is correctly valued for all land characteristics and all neighbourhoods.
5. Develop a market-based depreciation factor.
6. Calibrate the building values using the cost data provided.
7. Make market-based adjustments to the building values to ensure that all building characteristics are correctly valued.
8. When building value adjustments are completed, add the land and building values and test for valuation level across neighbourhoods.
9. Make any neighbourhood adjustments necessary.
10. Fully test the model and make any adjustments necessary.
11. State your conclusion as to model quality. Refer to the IAAO standards for homogeneous neighbourhoods.

Syntax Files

Students working in SPSS must include a syntax file (or files) with their Project submission. The syntax files should contain ONLY the transformations that were used. The syntax file(s) will help the marker quickly identify any problems and suggest corrective actions. This is vitally important because Project 1 will be used as a significant portion of the Case Study Examination – corrections will need to be made so that you do not lose marks on the exam for the same errors in your Project 1 submission.

Project Grading

There are two key learning outcomes for Project 1: (1) the preparation of an accurate and relevant valuation model; and, (2) the preparation of a comprehensive and informative report that explains what you did and why and discusses the practical importance of your model. To be successful in this project, you must complete BOTH of these aspects. While completing a good model is important, this outcome is considered secondary to demonstrating a good understanding of model building using regression analysis.

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PROJECT 1, continued

The tutor for this course has prepared a marking worksheet for Project 1 and you can find a copy of this under “Online Readings” on the BUSI 444 Course Resources web page. You are advised to review this worksheet closely in preparing your project submission, as it highlights the important points your tutor will be looking for when grading Project 1.

Marks for Project 1 will be allocated as follows:

	Mark Value
Report Introduction and Structure	10
Time Adjustment	10
Land Value: Model Development and Testing	25
Cost Model:	
Depreciation	15
Building Value Calculation and Testing	20
Neighbourhood Adjustments	10
Final Analysis and Conclusions	10
Total	100

The marker may also identify a “major fault” that results in a failing grade for the project regardless of the grade indicated from the sections above. A major fault would typically be an error that is of a degree of seriousness that the entire modeling exercise is undermined as a demonstration of competency. As noted previously, Project 1 will form a significant part of the Case Study Examination. As such, refer to your Project 1 results and the marking worksheet completed by the marker to assist you in preparing for the Case Study Examination.

Viewing Project 1 Grades

For Project 1, if you wish to view your grade on the Assignment History webpage, please note that Project 1 is labelled “Assignment 11”. Once your grade is posted you can view the feedback for your submission online. Because submissions will be individual, there is no answer guide for this project.

Note from the Tutor

The overall intent of this project is to have you apply the course concepts in a practical situation. This is a very challenging exercise and you will likely experience some difficulties along the way. The model is a crucial step towards completing this project, but it is the accompanying analysis that is particularly important. If you have a mistake in your model that means your conclusions are significantly off, you can still do fine on the project if your overall procedures are sound and your report documents these adequately. It is more important for you to have the best overall report you can, than to worry too much about having every aspect of your model perfect. So, try not to obsess about the details – keep the big picture in mind! Trust me, as difficult as it may seem now, when you have finished this project and course, you will be amazed at how much you have learned. Best wishes.

PROJECT 2: SALES-BASED MODELING CASE STUDY

See page **Projects.1** for details on project options and general expectations. The two projects together will form a comprehensive analysis.

Total Marks for Project 1: 20% of Course Grade



INTRODUCTION

This project and Project 1 require students to create valuation models based on a database of property sales. This project focuses on sales-based modeling and is a continuation of Project 1 to complete the residential valuation process. As such students must make use of cost information given in Project 1 that is relevant to the market model as well as any variables developed in Project 1 that are of use. Some of these have been provided in the project database. It is highly recommended that you do not “start from scratch” for Project 2 as there is information out of Project 1 that will be very useful here.

There are several learning outcomes for Project 2:

- Describe and state using standard mathematic symbols, an appropriate general model to use as an additive linear regression model;
- Review the available variables in the database using all analytic techniques covered in the course to select the variables suitable for this model type and transform variables as necessary;
- Specify and calibrate the model using an additive linear regression technique;
- Completely test, adjust, and evaluate the model for use; and
- State your conclusions as to model quality and provide recommendations to the Assessor of the City of Marilu for any necessary action which should be undertaken before using the model to value these neighbourhoods for the 2017 assessment roll.

Note from the Marker

This is the most important project in this course with respect to demonstrating an understanding of the principles covered for regression model building. In this project, there is both a requirement to carry out required calculations accurately and to explain your rationale for the work done. In your grade for this project, 60% will be based on your analysis, while 40% will be based on your description of the process used.

Length of Report

Your report must describe the processes you used, demonstrating your understanding of mass appraisal theory and knowledge of how to develop a sound, practical valuation model. However, your report should be presented in the format of an actual report as much as possible. The only statistics that should be included in the body of the report are those necessary to prove the validity of the opinion being expressed. The full detail reports should only be included as appendices to the main report. For example, if the mean and standard deviation obtained from a descriptive statistics report are included in your report, these should be followed by a reference to the full report (i.e., “Full details of this report may be found in the descriptive statistics report contained on page 3 of Appendix A”).

Your report should contain no more than 25 pages of report detail and will be supported by an appendix which contains all statistical reports that were relied on to justify any decisions or reach any conclusions. The appendix should be organized and indexed and may contain, on average, approximately 15 to 30 pages. Note that these are only rough limits and that you should provide all relevant information, i.e., anything which you used to substantiate decisions or adjustments. Note also that a very well-written and organized report could require significantly less than these numbers of pages.

The table below summarizes the suggested size of submissions and the maximum allowable pages. Under no circumstances should you need more than the stated maximum number of pages for your report. If your submission exceeds these limits, EITHER: 1) your assignment will be unmarked and you will receive a request for you to resubmit your assignment within the page length requirements; OR 2) the additional pages will not be read by the marker and marks will be deducted for their inclusion.

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PROJECT 2, continued

	Report	Appendices
Suggested	10 to 15 pages	15 pages
Maximum	25 pages	30 pages



Helpful Hint

For statistical output, be careful to **ONLY** provide the charts and tables that support your discussion – **DO NOT** overwhelm the marker with pages and pages of irrelevant output, as this will be considered a fault and marks deducted. A good rule to follow for statistical output is that you should not have it in your appendix unless you referred to it in your report – if it did not warrant mentioning in your report, then why is it relevant enough to make the marker wade through it? For most regression analysis reports, only the following tables are required: Model Summary, ANOVA, Coefficients (including Tolerance and VIF statistics).

The marker must be able to follow the logic of your analysis, and so must be able to see the transformations you made. Those working in SPSS must include syntax files showing all transformations made. Ideally, you should provide an electronic copy of any relevant output files so that the marker can review your work if necessary. Alternately the transformation syntax statements can be included as part of the appendices. It is especially important to include enough ancillary information so that the marker can make comments and suggestions for improving or correcting your Project because later in the course you will use a corrected version of Project 2 as part of your Case Study Examination.

Students are reminded that, in addition to a practical demonstration of mass appraisal processes, this is an examination of the underlying theory. You must ensure that you provide sufficient explanation of the process and procedures in order to demonstrate your knowledge of sound mass appraisal theory. While solid results are important, a good percentage of your grade is based on the explanation provided.

Project 2 – Sales-Based Modeling Case Study

In this project, you must develop and test a sales-based additive multiple regression model to appraise improved residential properties in Marilu City. Marilu City uses a full market value assessment base, so your model values must reflect the full actual value of the fee simple interest in the properties.

Your report must describe your results and the procedures used to obtain these results. The report should provide the Assessor of the Marilu City with a mass appraisal model which could be applied to value all improved residential properties in Marilu City. The valuation date is July 1, 2017 and the assessment roll is to be presented to Marilu City December 31, 2017. Your report will be presented to the Assessor on November 30, 2017. As a part of this exercise, you are expected to also fully critique the model, the results, and their applicability.

Database Description

This modeling exercise continues from Project 1, so you should ideally continue with the data from there. Alternatively, if you want a fresh start with new data you may instead download the database “MariluProject2_YEARTERM”, found on the BUSI 444 Course Resources webpage in the same location as the other database files for the lessons. The file is available in SPSS format.

The database contains 63 existing variables. There are 400 or more sales included in the database (each term has a slightly different database provided). The description of each variable may be viewed in the “Variable View” in SPSS.

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PROJECT 2, continued

For Project 2, you are expected to make all possible use of the information provided in Project 1, especially for carports and garages. Make appropriate transformations of the garage type, area, and quality factors given for garages, and also the area and quality information given for carports. The last six variables in the Marilu Project 2 database have been provided in the event that these values were not accurately calculated in Project 1. You can use these variables without an explanation of their origin as they are given as part of this Project's information.

Please note: students are recommended to continue with their existing Project 1 database, NOT use this Project2 database. The Case Study Examination will require joining Project 1 and Project 2 into a one continuous and comprehensive report based on your own data and calculations. If you use the new Project2 database here, for the exam you will have to return to your Project 1 database and re-create the Project 2 variables. This may be time-consuming for you later.

Case Study Examination and Databases

The final examination for this course is a take-home comprehensive case study that is based on Projects 1 and 2. You will be required to combine Projects 1 and 2 into a single report. This will require correcting your work based on the marker's feedback and creating a smooth transition between the two Projects (that is, part of the exam will be the successful joining of the two Projects). If you decide to use the Project 2 database discussed above, rather than continuing with your original Project 1 database, please note that for exam purposes, you will have to return to your Project 1 database and make all the necessary transformations. You are not allowed to use the Project 2 database provided above for your exam. This is because it contains a number of variables that you have to produce yourself, and will be tested on, to establish a smooth transition between Project 1 and Project 2 for the exam. As such, your results from Project 1 must feed into your Project 2 analysis.

Modeling Steps

The following steps must be completed in building your sales-based model:

1. Describe an appropriate general model to use as an additive linear regression model. This model must be stated with the standard mathematical symbols used in the lessons.
2. Review the variables in the database and identify those which are suitable to use as independent variables for the type of model defined in Step 1.
3. Examine the potential independent variables using descriptive statistics, graphical analysis, cross tabulation tables, and correlation analysis, looking for relationships with each other and with the dependent variable.
4. (a) Create any transformations necessary to make variables suitable for the chosen model structure.
(b) Create any additional transformations required to remove problems of collinearity identified in Step 3.
5. Repeat Step 3 with the new variables.
6. Run a number of initial models and list a final group of variables for calibration.
7. Calibrate the model using additive linear regression.
8. Completely test and evaluate the model for use.
9. State your conclusions as to model quality and provide recommendations for any necessary action which should be undertaken before using the model to value Marilu City properties.

Syntax Files

Students working in SPSS must include a syntax file or files with their Project submission. The syntax file(s) will help the marker quickly identify any problems and suggest corrective actions. This is vitally important because Project 2 will be used as a significant portion of the Case Study Examination – corrections will need to be made so that you do not lose marks on the exam for the same errors in your Project 2 submission.

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PROJECT 2, continued

Project Grading

There are two key learning outcomes for Project 2: (1) the preparation of an accurate and relevant valuation model; and (2) the preparation of a comprehensive and informative report that explains what you did and why and discusses the practical importance of your model. To be successful in this project, you must complete BOTH of these aspects. While completing a good model is important, this outcome is considered secondary to demonstrating a good understanding of model building using regression analysis.

The tutor for this course has prepared a marking worksheet for Project 2 and you can find a copy of this under “Online Readings” on the BUSI 444 Course Resources webpage. You are advised to review this worksheet closely in preparing your project submission, as it highlights the important points your tutor will be looking for when grading Project 2.

Marks for Project 2 will be allocated as follows:

	Mark Value
Report Introduction and Structure	10
Data Analysis	20
Transformations	15
Variable Selection	25
Model Calibration	10
Final Analysis and Conclusions	<u>20</u>
Total	100

The marker may also identify a “major fault” which results in a failing grade for the project regardless of the grade indicated from the sections above. A major fault would typically be an error that is of a degree of seriousness that the entire modeling exercise is undermined as a demonstration of competency. For example, inclusion of multicollinearity in final model or lack of recognition of multicollinearity in the model development process is a major fault. As noted previously Project 2 will be used again as a significant part of the Case Study Examination. As such, refer to your Project 2 results and the marking worksheet completed by the marker to assist you in preparing for the Case Study Examination.

Viewing Project 2 Grades

For Project 2, if you wish to view your grade on the Assignment History webpage, please note that Project 2 is labelled “Assignment 12”. Once your grade is posted you can view the feedback for your submission online. Because submissions will be individual, there is no answer guide for this project.

PROJECT 3: INCOME MODELING CASE STUDY

See page **Projects.1** for details on project options and general expectations. The two projects together will form a comprehensive analysis.

Total Marks for Project 1: 10% of Course Grade



Introduction

To continue with the mass appraisal focus of this course, the final Project will require building a linear MRA model based on income data. The procedure will be similar to that carried out in Lesson 7. You will select an appropriate unit of measure for creating an income-based valuation model, and then build the model using the selected unit of measure as the dependent variable. Each term the data may change, but the project requirements and marking guidelines, outlined below, will remain the same. The database as many as 100 records with up to 15 variables to choose from for your model. You may find further details for Project 3 under “Online Readings” on the Course Resources website.

Database Description

Visit the Course Resources website to download the Project 3 database and the associated project description.

Modeling Steps

The following steps must be completed in building your income model:

- A. Describe an appropriate general model to use as an additive linear regression model for an income approach application. Examine the data provided and determine what is the best market value indicator for this database: gross income multiplier, sale price per unit, or sale price per square foot. This will be the dependent variable to use in your model. Explain your conclusion.
- B. Create an additive multiple regression model to estimate the unit of measure you selected in Step A above. Describe the steps you followed to create your final model and show all pertinent output. In addition, you are also expected to briefly test and evaluate the model, and state your conclusions as to the quality of the model.

NOTE: Considerable caution should be exercised in the variable selection for the model in this question. It is extremely important that the variables to be used in the model are independent of each other and of the dependent variable selected. For example, if the dependent variable chosen is sale price per square foot, the sale price and square foot variables would be completely correlated with SP_SQFT and could not be used in a regression model with SP_SQFT as the dependent variable. In fact, any variable that was created using sale price is not a suitable independent variable for use in this model. Since this model is simply a preliminary step in developing value estimates for income-producing properties, using a variable such as PRICE, or any variable which requires price for its development, will render the model useless for valuation of properties that have not sold.

Following are the general steps required in creating an additive multiple regression model. They are repeated here as a reminder, but it is expected that these steps will be somewhat abbreviated in this project.

1. Describe an appropriate general model to use as an additive linear regression model for an income approach application.
2. Review the variables in the database and identify those which are suitable to use as independent variables for the type of model defined in Step 1.
3. Examine the potential independent variables using descriptive statistics, graphical analysis, cross tabs, and correlation analysis, looking for relationships with each other and with the dependent variable.
4. (a) Create any transformations necessary to make variables suitable for the chosen model structure.
(b) Create any additional transformations required to remove problems of collinearity identified in Step 3.

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PROJECT 3, continued

5. Repeat Step 3 with the new variables.
6. Run a number of initial models and list a final group of variables for calibration.
7. Calibrate the model using additive linear regression.
8. Completely test and evaluate the model for use.
9. State your conclusions as to model quality and provide recommendations for any necessary action which should be undertaken before using the model to value properties outside the database.

Length of Report

Your report must describe the processes you used, demonstrating your understanding of mass appraisal theory and knowledge of how to develop a sound, practical valuation model. However, your submission should be presented in the format of an actual report as much as possible. The only statistics that should be included in the body of the report are those necessary to prove the validity of the opinion being expressed. The full detail graphs and tables should only be included as appendices to the main report. For example, if the mean and standard deviation obtained from a descriptive statistics report are included in your report, these should be followed by a reference to the full report (i.e., "Full details of this report may be found in the descriptive statistics report contained on page 3 of Appendix A").

Your report should contain no more than 15 pages of report detail and will be supported by an appendix that contains all statistical information you relied on to justify any decisions or reach any conclusions. The appendix should be organized and indexed and may contain, on average, approximately 10 to 12 pages. Note that these are only rough limits and that you should provide all relevant information, i.e., anything which you used to substantiate decisions or adjustments. Note also that a very well-written and organized report could require significantly less than these numbers of pages.

The table below summarizes the suggested size of submissions and the maximum allowable pages. Under no circumstances should you need more than the stated maximum number of pages for your report. If your submission exceeds these limits, EITHER: 1) your assignment will be unmarked and you will receive a request for you to resubmit your assignment within the page length requirements; OR 2) the additional pages will not be read by the marker and marks will be deducted for their inclusion.

	Report	Appendices
Suggested	10 to 12 pages	10 pages
Maximum	15 pages	20 pages

Syntax Files

Students working in SPSS must include a syntax file or files with their Project submission. The syntax file(s) will help the marker quickly identify any problems and suggest corrective actions.

Project Grading

There are two key learning outcomes for Project 3: (1) the preparation of an accurate and relevant income valuation model; and (2) the preparation of a comprehensive and informative report that explains what you did and why and discusses the practical importance of your model. To be successful in this project, you must complete BOTH of these aspects. While completing a good model is important, this outcome is considered secondary to demonstrating a good understanding of model building using regression analysis.

The tutor for this course has prepared a marking worksheet for Project 3 and you can find a copy of this under "Online Readings" on the BUSI 444 Course Resources webpage. You are advised to review this worksheet closely in preparing your project submission, as it highlights the important points your tutor will be looking for when grading Project 3.

Project 3 continues on the following page

PROJECT 3, continued

Marks for Project 3 will be allocated as follows:

	Mark Value
Report Introduction and Structure	10
Data Analysis	15
Transformations	10
Variable Selection	25
Model Calibration	20
Final Analysis and Conclusions	<u>20</u>
Total	100

The marker may also identify a “major fault” which results in a failing grade for the project regardless of the grade indicated from the sections above. A major fault would typically be an error that is of a degree of seriousness that the entire modeling exercise is undermined as a demonstration of competency. For example, inclusion of multicollinearity in final model or lack of recognition of multicollinearity in the model development process is a major fault.

Viewing Project 3 Grades

For Project 3, if you wish to view your grade on the Assignment History webpage, please note that Project 3 is labelled “Assignment 13”. Once your grade is posted you can view the feedback for your submission online. Because submissions will be individual, there is no answer guide for this project.

